

# BIOL-8 Exam 02

College of the Redwoods, Del Norte

Instructor: Dr. Daniel Friedman

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Modules 07-10: Genetics, Cell Division, Tissues & Inheritance

**Date:** Thursday, March 5, 2026

**Time:** 5:30 PM - 8:40 PM

**Total Points:** 50

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### Part A: Multiple Choice (30 points)

*Select the best answer for each question. Each question is worth 1 point.*

#### Module 07: Genetics & DNA

1. The "central dogma" describes the flow of genetic information as:

- A) Protein → RNA → DNA
- B) DNA → RNA → Protein
- C) RNA → DNA → Protein
- D) DNA → Protein → RNA

2. A codon is made up of how many bases?

- A) One
- B) Two
- C) Three
- D) Four

3. What is the product of transcription?

- A) A copy of DNA
- B) mRNA
- C) A protein
- D) tRNA

4. A change (mutation) in DNA:

- A) Always changes the protein
- B) Sometimes changes the protein and sometimes does not
- C) Never changes the protein
- D) Always kills the cell

5. DNA replication is called "semiconservative" because each new DNA molecule contains:

- A) Only newly made strands
- B) One original strand and one new strand
- C) Only the template strand
- D) Two completely new strands

## **Module 08: Cell Division**

1. The main purpose of mitosis is:

- A) Making sex cells (gametes)
- B) Growth and repair
- C) Reducing chromosome number
- D) Creating genetic variation

2. DNA is copied during which phase of the cell cycle?

- A) G1 phase
- B) S phase
- C) G2 phase
- D) M phase

3. Which process produces two genetically identical cells?

- A) Meiosis I
- B) Meiosis II
- C) Mitosis
- D) Fertilization

4. At the end of meiosis, human sex cells contain:

- A) 46 chromosomes
- B) 23 pairs of chromosomes
- C) 23 chromosomes
- D) 92 chromosomes

5. How many cells are produced at the end of meiosis?

- A) One
- B) Two
- C) Three
- D) Four

## **Module 09: Tissues**

1. Which tissue type is specialized for contraction and movement?

- A) Epithelial
- B) Connective
- C) Muscle
- D) Nervous

2. Which organ system breaks down food into nutrients the body can use?

- A) Respiratory system
- B) Nervous system
- C) Digestive system
- D) Endocrine system

3. An organ like the stomach contains multiple tissue types working together. Which of the following is a tissue type found in the stomach wall?

- A) Bone tissue
- B) Muscle tissue
- C) Cartilage
- D) Gonads

4. The functional cells of nervous tissue are called:

- A) Fibroblasts
- B) Osteocytes
- C) Chondrocytes
- D) Neurons

5. The three types of muscle tissue are:

- A) Striated, smooth, and rough
- B) Skeletal, cardiac, and smooth
- C) Voluntary, involuntary, and mixed
- D) Fast-twitch, slow-twitch, and intermediate

6. Which organ system is responsible for transporting oxygen and nutrients throughout the body?

- A) Digestive system
- B) Respiratory system
- C) Circulatory system
- D) Skeletal system

7. Which type of muscle is under voluntary (conscious) control?

- A) Smooth muscle
- B) Cardiac muscle
- C) Skeletal muscle
- D) All muscle types are voluntary

## Module 10: Inheritance

1. When two heterozygous parents ( $Aa \times Aa$ ) are crossed, the expected ratio of genotypes ( $AA : Aa : aa$ ) is:

- A) 1:1
- B) 3:1
- C) 1:2:1
- D) 4:0

2. If both parents are carriers of a recessive trait ( $Aa \times Aa$ ), what is the chance their child will show the recessive trait?

- A) 0%
- B) 25%
- C) 50%
- D) 75%

3. A red flower ( $RR$ ) is crossed with a white flower ( $WW$ ). All offspring are pink. This is an example of:

- A) Complete dominance
- B) Incomplete dominance
- C) Codominance
- D) Polygenic inheritance

4. Human blood type is an example of:

- A) Simple dominance
- B) Incomplete dominance

- C) Multiple alleles and codominance
- D) Polygenic inheritance

5. A "carrier" is a person who:

- A) Is homozygous dominant
- B) Is homozygous recessive and shows the trait
- C) Is heterozygous and does not show the recessive trait
- D) Has no alleles for the trait

6. If you cross two parents that are both heterozygous genotype for two different dominant traits, the offspring will show:

- A) Only one combination of traits
- B) Two combinations of traits
- C) Four combinations of traits
- D) No variation in traits

7. A pedigree shows two unaffected parents with an affected child. This trait is most likely:

- A) Autosomal dominant
- B) Autosomal recessive
- C) X-linked dominant
- D) Y-linked

8. Males are more often affected by X-linked recessive conditions because:

- A) Males have two X chromosomes
- B) Males have one X, so one recessive allele is enough to cause the condition
- C) Females have a protective Y chromosome
- D) X-linked traits only come from fathers

9. A carrier mother ( $X^c X$ ) and a father with normal vision ( $X Y$ ) have sons. What fraction of their sons will be color blind?

- A) None
- B) One-quarter

- C) One-half
- D) All of them

10. The Law of Independent Assortment states that:

- A) Alleles always stay together
- B) Genes on different chromosomes sort independently during meiosis
- C) Dominant alleles always mask recessive alleles
- D) Traits always skip a generation

11. A father with blood type A (genotype AO) and a mother with blood type B (genotype BO) could have a child with blood type:

- A) A only
- B) B only
- C) AB only
- D) A, B, AB, or O

12. A genetic trait that requires two copies of the recessive allele (one from each parent) to appear is called:

- A) Autosomal dominant
- B) Autosomal recessive
- C) Sex-linked
- D) Codominant

13. Which of the following is true about sex-linked inheritance?

- A) Fathers pass X-linked traits to their sons
  - B) Mothers can pass X-linked recessive traits to their sons
  - C) X-linked traits affect males and females equally
  - D) Y-linked traits are more common than X-linked traits
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## Part B: Fill in the Blank (11 points)

Complete each statement with the correct term from the word bank. Each blank is worth 1 point.

**Word Bank:** *haploid, diploid, meiosis, mitosis, gametes, epithelial, connective, muscle, genotype, phenotype, dominant*

1. A cell with two complete sets of chromosomes is described as \_\_\_\_\_.
  2. Sex cells (sperm and eggs) are also called \_\_\_\_\_.
  3. A cell with only one set of chromosomes is described as \_\_\_\_\_.
  4. The type of cell division that produces body cells for growth and repair is \_\_\_\_\_.
  5. The type of cell division that produces sex cells is \_\_\_\_\_.
  6. The tissue type that covers body surfaces and lines body cavities is \_\_\_\_\_.
  7. Blood and bone are classified as \_\_\_\_\_ tissue.
  8. Skeletal, cardiac, and smooth are the three types of \_\_\_\_\_ tissue.
  9. An organism's genetic makeup (e.g., Aa) is its \_\_\_\_\_.
  10. An organism's observable characteristics (e.g., brown eyes) describe its \_\_\_\_\_.
  11. An allele that masks the expression of another allele is called \_\_\_\_\_.
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## Part C: Free Response (9 points)

Choose *THREE* of the following five questions. Answer in the provided spaces below. You may write, draw, or both. Each response is worth 3 points.

1. **Mitosis vs. Meiosis.** Compare mitosis and meiosis. Include the purpose of each, how many cells are produced, and whether the resulting cells are diploid or haploid.



2. **Blood Type Genetics.** A father has blood type A (genotype AO) and a mother has blood type B (genotype BO). Draw a Punnett square and list all possible blood types of their children.
3. **Tissues and Function.** Pick two of the four tissue types (epithelial, connective, muscle, nervous). For each, name one specific example and explain how its structure helps it do its job.
4. **From DNA to Protein.** Describe the two main steps that convert the information in DNA into a protein. Name each step and where in the cell it happens.
5. **X-linked Inheritance.** A woman is a carrier for an X-linked recessive trait. Her husband is unaffected. Explain why their sons are more likely to be affected than their daughters.

**Response 1**

**Question Number:** \_\_\_\_\_

**Response 2**

**Question Number:** \_\_\_\_\_

**Response 3**

**Question Number:** \_\_\_\_\_

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**Feedback (Optional)**

A. Is there anything else you would like to share (e.g., questions, comments, or notes for the professor)?

B. What percentage grade do you estimate you earned on this exam? How confident are you in that estimate (1–10)?

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*End of Exam 02*